Launch Control XL 3 — Synth Kick Board (lab/kick)

The synth board with a bass drum as the source. Column 1: button 1 = kick (sine kick or a drum machine), button 2 = rhythm.

At start every stage is off (raw kick). Button row 2 switches each stage on/off. FM, Drop and Noise only affect the sine kick.

The status box at the bottom right shows all values live, with units. Faders jump from their start value to their real position on first movement.

Knob Fader Button unused

Mode 16 — Synth Kick Board

	1 Kick	2 Envelope	3 Filter	4 Filter env.	5 FM	6 Pitch	7 Drive	8 Space
Row 1	CC 13 **Click** 0–1: short noise click on every kick (all kicks): attack, audible on small speakers.	CC 14 **Attack** 1 ms–1 s: fade-in. Anything above a few ms softens the click away.	CC 15 **Resonance** 0–20: boosts the cutoff region; on a kick a tonal, booming ring.	CC 16 **Filter attack** 1 ms–1 s: how fast the filter opens.	CC 17 **FM decay** 10 ms–2 s: how fast the FM overtones fade.	CC 18 **Pitch decay** 10 ms–1 s: how fast the pitch falls.	CC 19 **Level after** 0.1–1: bring the volume back down after drive.	CC 20 **Room size** 0.5–8: small room … cathedral.
Row 2	CC 21 **Drop** 0–48 semitones: built-in pitch drop of the sine kick (punch). Samples have theirs baked in.	CC 22 **Decay** 10 ms–2 s: length of the tail (with sustain 0). The main kick-length control.	CC 23 **HPF** 20 Hz–2 kHz: high-pass. 30-50 Hz = tighter; higher = thin, breakdown kick.	CC 24 **Filter decay** 10 ms–2 s: how fast it closes. Short = bright click, dark body.	CC 25 **FM sustain** 0–1: 0 = only a strike at the start of the kick.	CC 26 **Vibrato rate** 0–12 Hz: pitch wobble of the tail.	CC 27 **Bitcrush** 16–2 bit: digital, 8-bit.	CC 28 **Delay** 0–0.8: volume of the echoes.
Row 3	CC 29 **Noise** 0–1: noise mixed into the sine kick: dirtier, more like a sampled kick.	CC 30 **Release** 10 ms–2 s: fade-out after the step ends.	CC 31 **HPF resonance** 0–20: resonance of the high-pass.	CC 32 **Filter sustain** 0–1: how far open it stays.	CC 33 **Detune** 0–5 % off the ratio: inharmonic, metallic.	CC 34 **Vibrato depth** 0–2 semitones.	CC 35 **Coarse** 1–32: lower sample rate, lo-fi sampler.	CC 36 **Feedback** 0–0.9: how many echoes.
Fader Level	CC 5 **Level** 0–1, overall volume.	CC 6 **Sustain** 0–1: level while held. 0 = the kick dies away after the decay.	CC 7 **Cutoff (LPF)** 40 Hz–12 kHz: low-pass, low = dull and soft (click gone), high = open.	CC 8 **Env amount** 0–6 octaves: the filter opens on each kick and closes again. Needs column 3 on.	CC 9 **FM amount** 0–8: sine kick only. Overtones: thud, metallic, growl.	CC 10 **Pitch env** 0–24 semitones: extra pitch drop at the start: more punch, high + short = „zap“.	CC 11 **Drive** 0–6: distortion; hard techno kicks are heavily driven.	CC 12 **Reverb** 0–1: amount of reverb. With a low LPF: the techno rumble.
Buttons 1	CC 37 **Kick** sine, 808 s/l, 909 A/B, 707, Linn, DMX, SP-1200.	CC 38 **Note length** ¼, ½, full, ×2 of the step (step = until the next kick).	CC 39 **Filter type** 12 dB, 24 dB, ladder (Moog).	CC 40 **Direction** up / down per kick.	CC 41 **Ratio** 1, 2, 3, 0.5, 1.414 (metallic).	CC 42 **Tune** ±0, +2, +5, -3, -5 semitones (samples: speed).	CC 43 **Distortion type** 9 types: scurve … chebyshev.	CC 44 **Delay time** 1/8, 3/16, 1/4, 1/16.
Buttons 2	CC 45 **Rhythm** 4/4 → Broken → Gallop → Halftime.	CC 46 **Envelope on/off** Off = the kick as it is.	CC 47 **Filter on/off**	CC 48 **Filter env on/off** Only works with filter on.	CC 49 **FM on/off** Sine kick only.	CC 50 **Pitch/vibrato on/off**	CC 51 **Drive on/off**	CC 52 **Space on/off**

Guide

How to use this board

At start every stage is off: you hear the raw kick. Step through the kicks (column 1, button 1), then switch one stage on at a time and hear what it does to the kick.

Anatomy of a kick

A kick has three parts: the click (the first few ms, makes it audible on small speakers), the punch (the pitch drop from high to low, 20-80 ms) and the body or tail (the low sine that rings out). The drum machines differ mostly in how long and how deep the body is and how hard the click is.

The machines

808: deep, boomy sine, long tail; hip hop, electro, trap. 909: punchier, with a click and a shorter body; the house and techno kick. 707, LinnDrum, DMX, SP-1200: sampled 80s kicks, shorter and tighter; electro, hip hop, breaks. Sine kick: fully synthetic, every part adjustable (Drop, Noise, FM).

Shaping a kick

Envelope decay = length of the tail; short for fast techno, long for deep or 808 styles. Low-pass = darker, removes the click. High-pass at 30-50 Hz = tighter, less rumble. Pitch envelope = extra punch on top of the sample. Drive = louder, harder (hard techno kicks are distorted). Space = rumble (lots of reverb, then a low-pass).

Tuning

A kick has a pitch. Tune it to the key of the track (here A minor): a kick that clashes with the bass makes the low end muddy.

Recipes

Sound	Settings	Where in techno / house
Hard techno kick	909, drive hard 3-5, Level after down, envelope D ~200 ms S 0, filter 24 dB ~3 kHz	hard and industrial techno
Long 808	808 lang, envelope off, tune -3 or ±0, drive scurve 1	hip hop, trap, electro
Tight house kick	909 A, envelope D ~150 ms S 0, HPF ~40 Hz, click 0.3	house, tech house
Rumble	909, space on, reverb 0.7, room size 6, filter LPF ~150 Hz	hypnotic techno: the reverb tail becomes a low drone
Zap kick	sine kick, pitch env on 24 HT, pitch decay ~20 ms, drop 0	electro, minimal
FM thud	sine kick, FM on, ratio 1, amount 2-3, FM sustain 0, FM decay ~30 ms	dark, metallic kicks
Lo-fi kick	SP-1200 or DMX, drive on, bitcrush ~8, coarse 4	breaks, lo-fi house